

# Step 5: Visualize — Design Record

CAPSTONE-01 | AI-Assisted Audit Control Framework · MCX AI Lab: Data Analysis with AI  
Dataset: `grc_risk_register.csv` — synthetic GRC risk register, 160 rows × 19 columns · snapshot date 30 Jun 2026

## 1 PURPOSE AND DELIVERABLE

This record documents how the visualization was designed — what was encoded, what was deliberately left out, and what was corrected during the build. It is the design rationale behind the artifact, not the artifact itself. The deliverable is **CAPSTONE-01\_GRC\_Risk\_Register\_Dashboard\_FINAL.html**: a single self-contained file, 63 KB, zero external dependencies, carrying two tabbed layers over one dataset — an **Executive Summary** for leadership and an **Auditor Detail** layer for the committee. Both layers are generated from the same verified figures, so they cannot drift from each other or from the Step 4 verdicts. Every claim admitted to this artifact passed Step 4; every claim Step 4 withdrew is absent by design.

## 2 DESIGN DECISIONS LOCKED

|                                     |                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two layers, not two documents       | One artifact, two tabs, one computation run. The Step 2 brief called for a ten-second leadership view over an exception-and-evidence layer; tabs deliver that without producing a second file that can fall out of sync. The executive layer states conclusions in plain language, the auditor layer proves them. Neither is a diluted version of the other. |
| Denominator on the face             | Each exhibit header carries its denominator as a literal fraction — 33/116, 33/33, 6/33. This puts the central Step 2 decision on the document rather than in a footnote. It is the most likely point of challenge in the room, and the artifact answers it before it is asked.                                                                              |
| Uncertainty is drawn, not footnoted | Every rate carries a Clopper-Pearson 95% interval rendered as a whisker across the bar. A reader can see two groups overlap without reading a p-value. This is the single most load-bearing encoding decision in the pack: it is what stops a 46.7% bar next to an 11.8% bar from reading as a finding.                                                      |
| Colour carries meaning              | Severity colour is reserved for exhibits where a difference is real. The null exhibit is drawn in a single neutral tone. A red-to-green scale there would imply a ranking the data cannot support, purely as a by-product of styling.                                                                                                                        |
| Plain language first                | Each exhibit leads with the conclusion in plain English and closes with a small <b>Statistical basis</b> line carrying the tests and p-values. Seven such notes appear in the auditor layer. Removing the statistics would make the pack undefendable; leading with them would make it unread.                                                               |
| Synthetic provenance stated         | The masthead states that the pack demonstrates a method on synthetic data and that findings describe this register, not an institution. Placed in the standfirst rather than the footer, because it changes how every figure below should be read.                                                                                                           |

## 3 WHAT WAS BUILT

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Executive layer      | Five KPI tiles (closed, open-not-due, past due now, ever missed deadline, median time past due); a bottom-line statement; Exhibit 1 book position as a stacked bar; Exhibit 2 control effectiveness as ranged bars; Exhibit 3 aging with exposure; Exhibit 4 the six largest past-due items; a five-point action block; and the supported / not-supported scope box.                                                                                                                                                                                                            |
| Auditor layer        | Section A population reconciliation and denominator policy, including the 34-item adjustment shown as an explicit accept / reject table and the Days_Overdue limitation with all three breach bases; Section B structural dependencies between tier, score and exposure; then Exhibits A1 to A6 — control effectiveness with intervals, the tier-stratified matrix, aging, the null-results set, tier with the SLA-window column, and the top-six table with its reconciliation. Eleven tables, seven statistical-basis notes, and the scope box restated with test statistics. |
| Encoding rules       | Bar length encodes rate in the effectiveness, source and tier exhibits, and dollars in the aging exhibit — stated in the axis label each time. Stacked-bar segment width encodes item count, labelled as such because the dollar split differs. No dual axes, no donuts, no encoding without a stated referent.                                                                                                                                                                                                                                                                 |
| Thin-base discipline | Groups under 20 items are marked thin, applied mechanically to all twenty occurrences. Step 3 had applied the same stated rule selectively, marking two groups and missing four; the rule is now enforced rather than stated.                                                                                                                                                                                                                                                                                                                                                   |
| Degradation          | Both layers are present in the markup and the tab bar is revealed by script, so with scripting blocked the artifact renders as one continuous document with a Part 2 divider. The auditor layer cannot silently disappear in a submission portal or document previewer. Print rules drop the tab bar, expand both layers and page-break between them.                                                                                                                                                                                                                           |

## 4 WHAT WAS NOT BUILT, AND WHY

The original specification called for a bar chart of overdue rate by department and a risk-tier donut. Neither was built. This section records the omissions deliberately, because an absent chart is invisible to a reader and therefore has to be argued for in writing.

|                 |                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CRITICAL</b> | <b>Department bar chart — not built.</b> Overdue rate by department ranges 16.7% to 42.9%, which charts well and means nothing: $\chi^2 p = 0.80$ on base populations of 6 to 20 items. A leader reading the tallest bar would ask a named department to explain a gap that is ordinary variation. The chart would generate an action the data cannot justify. |
| <b>CRITICAL</b> | <b>Risk-tier donut — not built.</b> Tier does not predict breach ( $p = 0.43$ ), and per Step 4 test 1 the apparent severity gradient is SLA-window construction. A donut would also encode a variable that is a deterministic banding of inherent score — three views of one underlying field presented as if independent.                                    |
| <b>MATERIAL</b> | <b>Owner-role, risk-domain and MAP-linkage cuts — not built.</b> $p = 0.33, 0.76$ and $0.99$ respectively. All five null dimensions are published instead as a single table with test statistics attached, so the absence of a finding is auditable rather than merely an unexplained gap in the pack.                                                         |
| <b>MATERIAL</b> | <b>Overrun ratio — retained but demoted.</b> Not removed, because a reviewer would ask why a standard metric is missing. Shown beside the SLA-window column that drives it, with the reader directed to median days past due. The exhibit states in terms that the ratio must not be cited.                                                                    |
| <b>WATCH</b>    | <b>Trend and cohort visuals — not built.</b> Excluded per Steps 1 and 2: cohort closure rate is flat by construction and recent cohorts are right-censored. A time series here would be a picture of the generator, and it would be the most persuasive and least true object in the pack.                                                                     |

## 5 CORRECTIONS MADE DURING THE BUILD

|                           |                                                                                                                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Headline corrected</b> | The on-time closure KPI read 85%, computed over all 117 closed items. Rebuilt on the 83 that faced a deadline: <b>78.3%</b> . The tile carries its denominator in the sublabel so the correction is visible rather than silent.                                                                                      |
| <b>Metric added</b>       | A fifth KPI tile was added for <b>ever missed deadline — 44.0%</b> , because the register zeroes the overdue counter on closure and 18 late closures were otherwise invisible. Currently-past-due and ever-breached now sit side by side; the band note explains that they answer different questions.               |
| <b>Exhibit added</b>      | The tier-stratified matrix (A2) was added to answer the confounding challenge visibly. Step 3 had asserted tier-independence without showing it. The matrix is published with a caution that no individual cell is a finding, and the conclusion drawn from the collapsed comparison beneath it.                     |
| <b>Exhibit demoted</b>    | Identification source moved from a lead finding to a null result, retitled accordingly, and redrawn in neutral colour with intervals. The “four times higher” comparison was removed from the lede.                                                                                                                  |
| <b>Lead reordered</b>     | “Not Tested behaves as Partially Effective” was promoted from a supporting callout to the lead. It rests on larger bases than the fourteen-fold spread, is statistically unambiguous, and carries a direct control implication. The fourteen-fold figure became supporting detail with its interval attached.        |
| <b>Layout defect</b>      | Prose blocks carried character-based maximum widths that truncated lines mid-page and broke alignment with the tables beneath them. Removed throughout; body text now spans the full column and aligns to the exhibits.                                                                                              |
| <b>Failure mode fixed</b> | The auditor layer initially started hidden and depended on script to appear. In any environment that blocks scripting the entire layer would have vanished with no indication — a reader would have seen a complete-looking document and assumed that was the whole submission. Inverted to progressive enhancement. |

6 WHAT THE VISUALIZATION DOES NOT CLAIM

SHOWN AND SUPPORTED

- Book position by state, count and exposure, reconciled to 160 items and \$147.0M.
- Control effectiveness against on-time closure, holding within every risk tier.
- “Not Tested” performing as “Partially Effective” — 42.1% against 42.4%.
- Aged-exposure concentration — 21 items beyond 180 days holding 94.6% of past-due dollars.
- Both breach bases, 28.4% current and 44.0% ever, with the 18 late closures disclosed.
- The six largest past-due items, reconciled against the tier total.

NOT CLAIMED BY THIS ARTIFACT

- Any performance difference between departments, owners, sources or domains.
- That risk tier predicts whether an item breaches.
- That exposure explains anything — it is banded from tier and shown as magnitude only.
- Any trend, improvement or cohort comparison.
- That control effectiveness reduces residual risk — Step 4 test 3 shows it does not.
- That any observed relationship reflects real-world control behaviour rather than dataset construction.

7 HANDOVER

|                  |                                                                                                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status           | Complete. All five method steps closed: Explore, Frame, Analyze, Verify, Visualize. The artifact carries no pending-review flag, because the review in Step 4 is complete and its withdrawals are reflected in the build rather than annotated on it.                                                                                                   |
| Reproducibility  | Every figure traces to a single pandas run over the source CSV; none transcribed by hand. The pack was authored natively rather than converted from a layout tool, so no rendering step can alter a number.                                                                                                                                             |
| Known constraint | The dataset is synthetic and three fields are not independent. The artifact demonstrates method and is not evidence about controls. Any reuse against a live register must repeat Step 1 profiling before any exhibit is trusted, because the denominator rule and the Days_Overdue limitation are properties of this file and may not hold in another. |
| Outstanding      | The recommended <b>Days_Late_At_Closure</b> schema change sits with the owner of the register, not with this analysis. It is the only recommendation in the five-step pack that changes a system rather than a document.                                                                                                                                |

**METHOD AND CONFIDENCE.** Confidence in every figure displayed is high — all were independently recomputed and reconciled during Step 4 before being admitted to the artifact. Confidence in the design decisions is stated rather than measured: the choice not to chart a null dimension is a judgement about how readers use charts, defended in section 4 rather than derived from the data. Confidence that any displayed relationship reflects real-world control behaviour is withheld, per Step 4 test 2. All rates carry Clopper-Pearson 95% intervals; groups under 20 items are marked thin without exception.

**SOURCE AND ACCOUNTABILITY.** Synthetic GRC risk register for portfolio demonstration; no production or employer data. Risk tier is a deterministic banding of inherent risk score and financial exposure is banded from tier; no exhibit treats them as independent. Artifact and design record produced with Claude (Opus 5) under human direction; every figure regenerated from source, none transcribed. Design decisions, omissions and corrections reviewed and owned by N. Fragoso. Prepared: 01 Aug 2026.